

Amna Jamal

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EDUCATION

Comsats University Islamabad

Bachelors of Computer Science

February, 2022 - February, 2026

CGPA: 3.44 /4.00

EXPERIENCE

Algoverse

AI Researcher

July, 2025 - Present

- Led my team on conducting mechanistic interpretability research on high-norm activations in Diffusion Transformers using causal ablation, probing, attention and MLP analysis to study their emergence and role in diffusion models. Became the First Author of the paper.
- Paper has recently been accepted at AAAI 2026 XAI4Science workshop
- Working on two more research projects on understanding the inner mechanisms of Diffusion Transformers.

Bluedot Impact

Technical AI Safety Researcher

May, 2026 - Present

- Working on a technical AI Safety research project

ResearchCircle

Founder, Campus Lead & Quantum Research Lead

September, 2025 - Present

- Founded a community dedicated at promoting a culture of research at undergraduate level via weekly circles of discussions, and workshops conducted by Researchers. I manage, and lead more than 600 students from different backgrounds e.g, Mathematics, Physics, Computer Science etc.
- I led the sub circle of Quantum & Advanced Computing from September 2025 till February 2026

AI Safety Research Lead

March, 2026 - Present

- I have started leading the AI Safety sub circle, with the aim of advocating for AI Safety in Pakistan. I facilitate discussions on different topics within technical AI safety.

Webrange Solutions

AI Intern

September, 2024 - October, 2025

- Developing an AI-powered platform that generates images and videos from multimodal inputs, applying generative modeling, data processing, and pipeline optimization techniques to handle complex visual data efficiently.

10Pearls

SWE Intern

June, 2025 - August, 2025

- Developed a full-stack contact management system in the Java stack, contributing to both development and testing.

SKILLS

- Programming Languages:** Python, Java, JavaScript, SQL, HTML, CSS, Linux
- Tools & Technologies:** PyTorch, Qiskit, Transformers, Hugging Face Diffusers, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib, Seaborn, Node.js, React, Express.js, MongoDB, MySQL, Git/GitHub, Matlab

PROJECTS

- High-norm Activations in Diffusion Transformers (Research)** - Conducted comprehensive analysis of diffusion transformers (DiT) by examining activation norms across attention and MLP layers, and visualizing them to study the role and emergence of high-norm activations. Paper accepted at AAAI 2026 XAI4Science workshop.
- Interpretability of Diffusion Transformers (Research)** - Currently working on a research project aimed at understanding and steering Diffusion Transformer using SAEs.
- Concept Unlearning in DiTs (Research)** - Working on concept unlearning via Steering Vectors in DiTs.
- AI-Powered Fashion Ad Generator** - Built a diffusion-based multimodal AI pipeline to generate realistic fashion ad images and videos from text and product image.
- RAG-based Chatbot** - Developed a Retrieval-Augmented Generation chatbot integrating LLMs with external knowledge sources for accurate, context-aware responses.

EXTRA-CURRICULAR

- Attended **Harvard Summer School** on particle physics
- Achieved **Quantum Excellence at IBM Qiskit Global Summer School 2025**
- Attended **5th Riazuddin Memorial School** on Quantum Computing
- Attended **Global Physics Summit 2025**.
- Member of the **American Physical Society** and **Google Developers Group (GDGOC)**.
- Participated in **hackathons**
- Served as **Peer Tutor** - assisted fellows with concepts in Differential Equations, and Linear Algebra.
- Took part in **Bluedot Impact Technical AI Safety**, **AGI Strategy**, and **Frontier AI Governance** programs.

INTERESTS

Generative AI, Mechanistic Interpretability, AI Safety, Quantum AI/ML, Particle Physics